



LED 테스트

CRZ Technology





Mango-M32F2 LED 테스트

LED pinmap

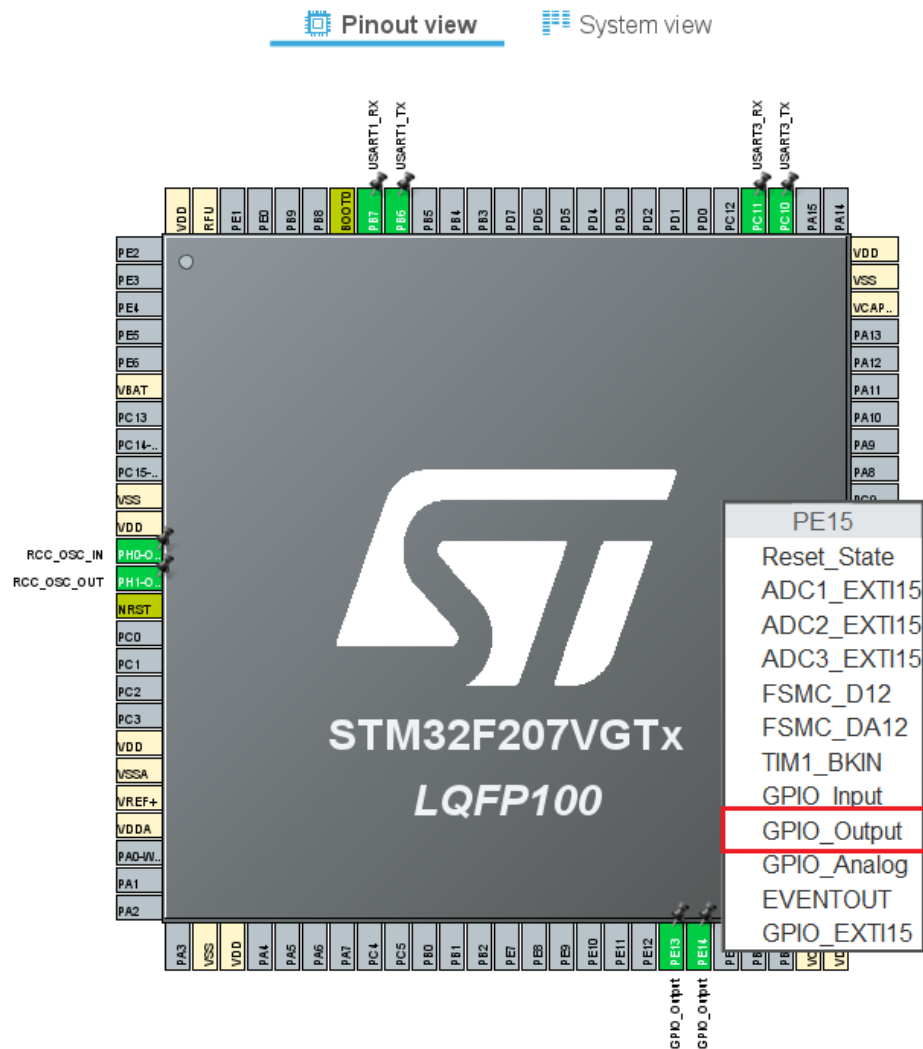
■ LED – PE13 / PE14 / PE15

PE12	44	PE13
PE13	45	PE14
PE14	46	PE15
PE15		

PE13	LED1
PE14	LED2
PE15	LED3

Pinout view

■ PE13 / PE14 / PE15 – GPIO_Output 선택.



GPIO 설정

- “System Core“ 클릭, “GPIO” 클릭.
- LED가 LOW이면 커지므로 초기값을 HIGH로 설정.

led_prj.ioc - Pinout & Configuration

Pinout & Configuration | Clock Configuration | Project Manager

Software Packs | Pinout

GPIO Mode and Configuration

Configuration

Group By Peripherals

GPIO | RCC | USART

Search Signals

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Pin Name	Signal on Pin	GPIO output le.	GPIO mode	GPIO Pull-up/	Maximum outp.	User Label	Modified
PE13	n/a	High	Output Push P...	No pull-up and ..	Low	LED1	☒
PE14	n/a	High	Output Push P...	No pull-up and ..	Low	LED2	☒
PE15	n/a	High	Output Push P...	No pull-up and ..	Low	LED3	☒

GPIO output level: High

GPIO mode: Output Push Pull

GPIO Pull-up/Pull-down: No pull-up and no pull-down

Maximum output speed: Low

User Label: LED1

LED Test Project

■ core/inc/main.h

```
59 /* Private defines -----*/
60 #define LED1_Pin GPIO_PIN_13
61 #define LED1_GPIO_Port GPIOE
62 #define LED2_Pin GPIO_PIN_14
63 #define LED2_GPIO_Port GPIOE
64 #define LED3_Pin GPIO_PIN_15
65 #define LED3_GPIO_Port GPIOE
```

LED Test Project

■ core/src/main.c 편집

```
103  /* USER CODE BEGIN 2 */
104  int count = 0;
105  printf("LED test\r\n");
106  /* USER CODE END 2 */
107
108  /* Infinite loop */
109  /* USER CODE BEGIN WHILE */
110  while (1)
111  {
112      HAL_GPIO_WritePin(LED1_GPIO_Port, LED1_Pin, !(count & 0x01));
113      HAL_GPIO_WritePin(LED2_GPIO_Port, LED2_Pin, !((count >> 1) & 0x01));
114      HAL_GPIO_WritePin(LED3_GPIO_Port, LED3_Pin, !((count >> 2) & 0x01));
115      count = (count + 1) % 8;
116      HAL_Delay(1000);
117      /* USER CODE END WHILE */
118
119      /* USER CODE BEGIN 3 */
120  }
121  /* USER CODE END 3 */
---
```

Test 결과 확인

- LED 3개 동작 확인.

Thank You

